VISHWANATH.S.A

Aspiring AI/Machine Learning Intern | Building End-to-End AI/ML Systems

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EXECUTIVE SUMMARY

CS undergraduate specializing in Machine Learning, with hands-on experience building and deploying end-to-end AI systems. Developed a fraud detection model on 6.3M+ transactions, a real-time traffic ETA prediction system, and a deployed drowsiness detection application using deep learning and computer vision. Seeking an AI/ML internship to design and ship scalable, real-world solutions.

SKILLS

Programming & Tools: Python, C, JS, Git
Data & Analysis: Pandas, NumPy, EDA, Data Cleaning, Data Visualization, Time-Series Processing
Machine Learning & AI: Logistic Regression, TF-IDF, XGBoost, Random Forest, Feature Engineering, MAE Evaluation, Deep Learning, Computer Vision
Frameworks & Libraries: PyTorch, Torchvision, OpenCV, Scikit-learn, Hugging Face
Databases: SQL, Firebase, Azure SQL

PROJECTS

Real-Time Drowsiness & Fatigue Detection System | Demo

- Built a real-time drowsiness detection system using Transfer Learning (MobileNetV2) trained on 2,900 images across 4 classes: Open, Closed, Yawn, and No Yawn.
- Applied data augmentation (flips, rotation, brightness jitter) to address limited dataset size and improve model generalization.
- Integrated OpenCV for live webcam inference with consecutive-frame alert logic — triggers drowsiness alarm only after 20+ closed/yawn frames to eliminate false positives.
- Deployed on Hugging Face Spaces with a Gradio interface — accessible as a live demo with image upload and real-time webcam support.

Fraud Detection using Machine Learning — Python, Pandas, Scikit-learn, EDA

- Built a fraud detection model on 6.36M financial transactions with only 0.13% fraud cases.
- Cleaned the dataset, handled structural zero balances, and kept meaningful high-value outliers.
- Engineered balance-difference features and analyzed fraud patterns in TRANSFER and CASH_OUT transactions.
- Trained balanced Logistic Regression and Random Forest models, achieving 98% fraud recall and 0.999 ROC-AUC with Random Forest.

Real-Time Traffic & Route ETA Forecasting using Machine Learning | Demo

- Built an ML pipeline on the METR-LA highway sensor dataset (34,000 time steps across 207 road segments).
- Converted multivariate traffic data into sliding time windows to predict segment speeds 15 minutes into the future.
- Trained independent XGBoost models per road segment and aggregated predicted ETAs to estimate end-to-end travel time for 30 km routes.

EDUCATION

Bachelor of Engineering in Computer Science and Engineering — **September 2023 - May 2027**
Kcg College Of Technology, Karapakkam, Tamil Nadu — Chennai, Tamil Nadu
CGPA: 7.9/10

ACHIEVEMENTS

- Conducted a hands-on 3-week workshop series for 30+ students covering cloud fundamentals, GitHub, static web hosting, storage accounts and serverless functions (Azure).